MOBILE SALES PERFORMANCE DASHBOARD

Business Analysis Report · Power BI · DAX · SQL · Data Modelling

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github.com/Ajay221301/Mobile_sales_dash

3,835 Transactions · 4 Years (2021–2024) · 19 Cities · 5 Brands

01 EXECUTIVE SUMMARY

This report delivers a comprehensive business intelligence analysis of mobile phone sales across **19 Indian cities, 5 major brands, and 4 years (2021–2024)**, covering 3,835 transactions worth ■76.9 crore total revenue. Built end-to-end from raw Excel data through **MySQL queries**, **Power Query transformation**, and an interactive 3-page **Power BI dashboard** with DAX time-intelligence (MTD, QTD, SPLY).

Six core business questions are answered: which cities and brands lead revenue, how performance trends monthly and annually, what payment behaviours exist, how satisfaction varies across segments, and where growth is declining.

TOTAL REVENUE
■76.9 Cr
2021 – 2024

TRANSACTIONS
3,835
Unique Records

UNITS SOLD
19,150
5 Brands

AVG PRICE
■40,114
Per Unit

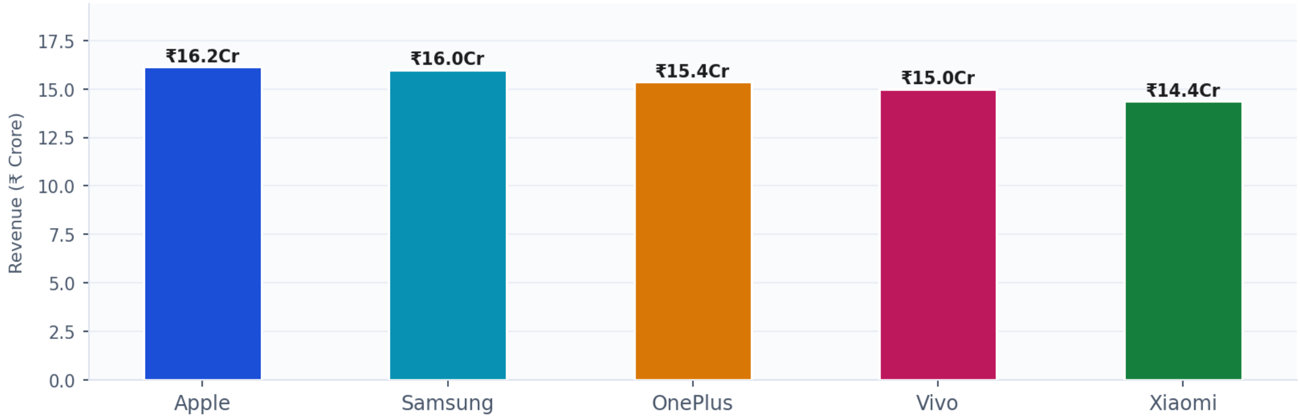
Year-on-Year Performance Snapshot

Year	Revenue	Transactions	YoY Growth	Signal
2021	■5.88 Cr	281	— Base —	■ Launch Year
2022	■26.20 Cr	1,250	**+345.4%**	▲ Breakout Growth
2023	■25.33 Cr	1,249	**–3.3%**	■ Stable Plateau
2024	■19.51 Cr	1,055	**–23.0%**	▼ Sharp Decline

02 BRAND & GEOGRAPHIC ANALYSIS

Revenue by Brand

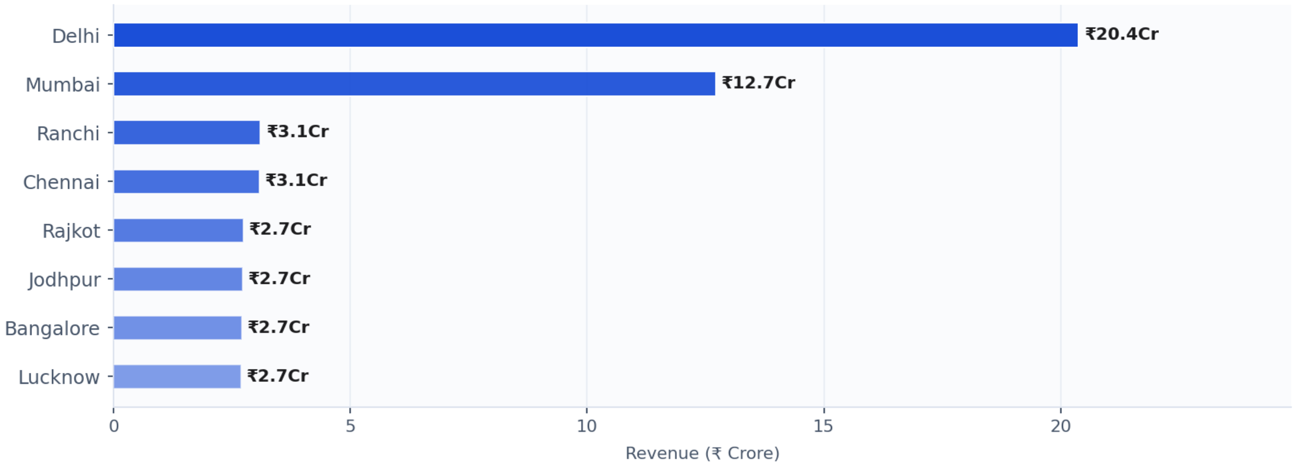
Which brand contributes most to total sales?



Apple leads at ■16.2 Cr (21.0%) with Samsung close behind at ■16.0 Cr (20.8%). All five brands compete within a tight ±12% range — no single dominant player, indicating a highly competitive and fragmented market.

Top 8 Cities by Revenue

Geographic revenue concentration



Delhi dominates at ■20.4 Cr (26.5% market share) — nearly 1.6× Mumbai's ■12.7 Cr (16.5%). The top two cities alone contribute 43% of all revenue, making them critical priority markets for inventory allocation and targeted campaigns.

01 Apple Leads

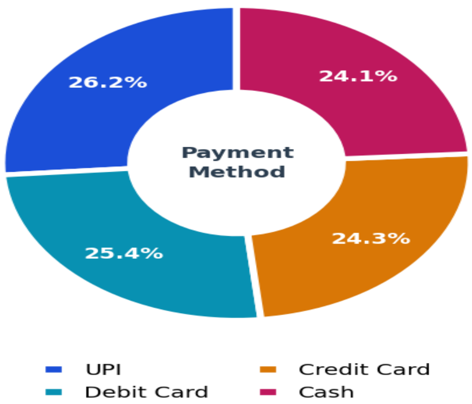
■16.2 Cr total. Highest avg price ~■55K+. Premium segment drives top-line performance.

02 Delhi Dominates

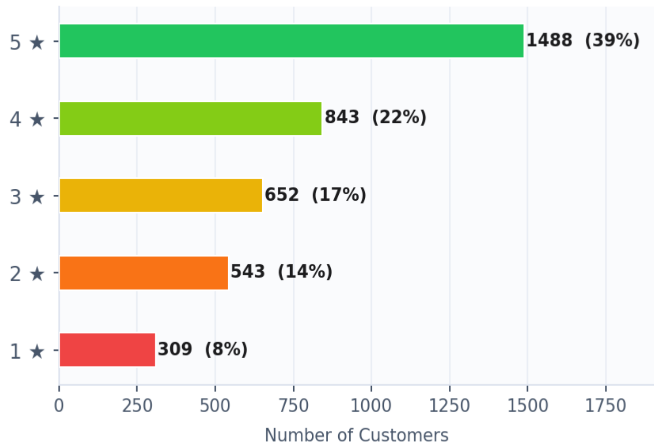
■20.4 Cr from Delhi alone. 3× third-placed city. #1 inventory & marketing priority.

03 PRODUCT, PAYMENT & CUSTOMER ANALYSIS

Payment Method Share
Revenue distribution



Customer Ratings
Distribution — 3,835 transactions



UPI leads at 26.4% revenue share — yet all four payment methods are nearly equal, reflecting diverse customer profiles. 38.8% of customers gave 5★ ratings (1,488 reviews); only 8.1% gave 1★ — signalling strong overall satisfaction across all brands.

Top 5 Mobile Models by Revenue
Which SKUs drive the most sales?

#	Mobile Model	Revenue	Revenue Share	Segment	Brand
1	**iPhone SE**	■5.96 Cr	7.7%	Budget-Premium	Apple
2	**OnePlus Nord**	■5.79 Cr	7.5%	Mid-Range	OnePlus
3	**Galaxy Note 20**	■5.60 Cr	7.3%	Mid-Range	Samsung
4	**Vivo Y51**	■5.48 Cr	7.1%	Budget-Mid	Vivo
5	**Galaxy S21**	■5.33 Cr	6.9%	Premium	Samsung

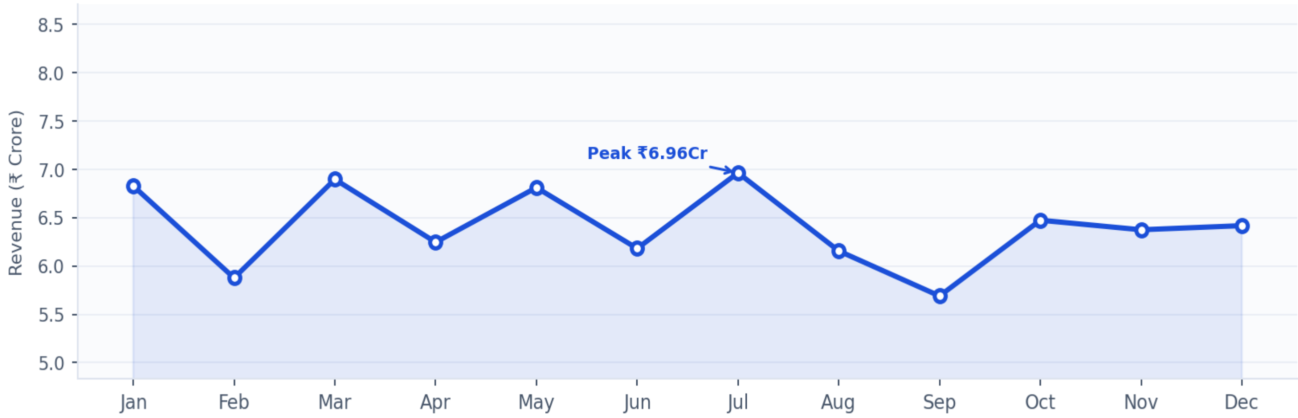
03 **UPI #1 Method**
26.4% txn share. Digital-first customers dominate all cities and age groups.

04 **iPhone SE Top**
■5.96 Cr. Apple budget-premium model outperforms all 14 other SKUs in the dataset.

04 TIME-SERIES TREND ANALYSIS

Monthly Revenue Trend

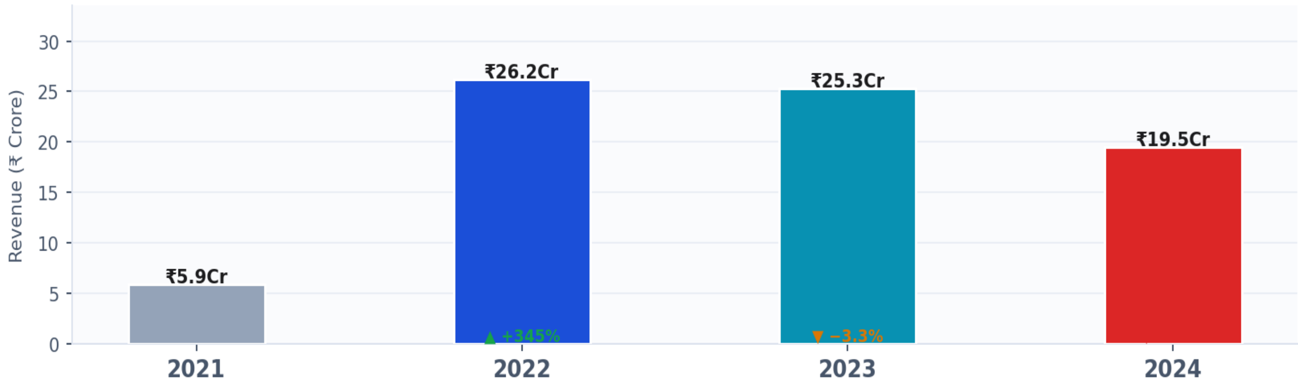
Aggregated across all 4 years — seasonal demand patterns



July is the consistent revenue peak at ■6.96 Cr across all years. September shows a recurring seasonal trough. March, May, and January are the next-strongest months — suggesting Q1 promotional cycles and mid-year product launches drive demand spikes.

Annual Revenue — Year-on-Year Comparison

Growth trajectory from 2021 to 2024



2022 was the breakout year (+345.4% YoY) as the business scaled from a partial launch in 2021. Revenue plateaued in 2023 (–3.3%) and fell sharply in 2024 (–23.0%), indicating possible market saturation, increased competition, or a data truncation effect for 2024.

05 **July = Peak Month**
Consistently highest revenue. Pre-load stock in late June; launch offers on July 1st.

06 **2024 Revenue Down**
–23% YoY is a material warning. Drill by city + brand to isolate the root cause.

05 SQL BUSINESS QUERIES

Six MySQL queries validate Power BI metrics independently and answer core business questions directly from the database — demonstrating end-to-end data fluency from raw SQL through visual BI reporting.

Q1 — Core KPIs: Revenue, Units & Transactions

MySQL

```sql
SELECT
SUM(Units_Sold * Price_Per_Unit) AS Total_Revenue,
SUM(Units_Sold) AS Total_Units_Sold,
COUNT(Transaction_ID) AS Total_Transactions,
AVG(Price_Per_Unit) AS Avg_Price_Per_Unit
FROM mobile_sales;
```

→ ■76.9 Cr total | 19,150 units | 3,835 transactions | ■40,114 avg price

Q2 — Brand Revenue Ranking with Market Share

MySQL

```sql
SELECT Brand,
ROUND(SUM(Units_Sold * Price_Per_Unit) / 1e7, 2) AS Revenue_Cr,
ROUND(SUM(Units_Sold * Price_Per_Unit) /
SUM(SUM(Units_Sold * Price_Per_Unit)) OVER() * 100, 1) AS Market_Share_Pct
FROM mobile_sales GROUP BY Brand ORDER BY Revenue_Cr DESC;
```

→ Apple 21.0% > Samsung 20.8% > OnePlus 20.0% > Vivo 19.5% > Xiaomi 18.7%

Q3 — Top Cities by Revenue & Geographic Concentration

MySQL

```sql
SELECT City,
ROUND(SUM(Units_Sold * Price_Per_Unit) / 1e7, 2) AS Revenue_Cr,
ROUND(SUM(Units_Sold * Price_Per_Unit) /
SUM(SUM(Units_Sold * Price_Per_Unit)) OVER() * 100, 1) AS Market_Share_Pct
FROM mobile_sales GROUP BY City ORDER BY Revenue_Cr DESC LIMIT 5;
```

→ Delhi 26.5% | Mumbai 16.5% | Ranchi 4.0% | Chennai 4.0% | Rajkot 3.6%

Q4 — Year-on-Year Growth using LAG Window Function

MySQL

```sql
SELECT Year,
ROUND(SUM(Units_Sold * Price_Per_Unit) / 1e7, 2) AS Revenue_Cr,
ROUND((SUM(Units_Sold * Price_Per_Unit) -
LAG(SUM(Units_Sold * Price_Per_Unit)) OVER (ORDER BY Year)) /
LAG(SUM(Units_Sold * Price_Per_Unit)) OVER (ORDER BY Year) * 100, 1) AS YoY_Pct
FROM mobile_sales GROUP BY Year ORDER BY Year;
```

→ 2022: +345.4% | 2023: –3.3% | 2024: –23.0%

Q5 — Payment Method Preference & Transaction Share

MySQL

```sql
SELECT Payment_Method, COUNT(*) AS Transactions,
ROUND(SUM(Units_Sold * Price_Per_Unit) / 1e7, 2) AS Revenue_Cr,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1) AS Txn_Share_Pct
FROM mobile_sales GROUP BY Payment_Method ORDER BY Revenue_Cr DESC;
```

→ UPI 26.4% | Debit Card 24.7% | Credit Card 24.7% | Cash 24.2%

Q6 — Customer Satisfaction by Brand (Avg Rating + 5★ Count)

MySQL

```
SELECT Brand,
ROUND(AVG(Customer_Ratings), 2) AS Avg_Rating,
COUNT(CASE WHEN Customer_Ratings = 5 THEN 1 END) AS Five_Star_Count,
ROUND(COUNT(CASE WHEN Customer_Ratings = 5 THEN 1 END)
* 100.0 / COUNT(*), 1) AS Five_Star_Pct
FROM mobile_sales GROUP BY Brand ORDER BY Avg_Rating DESC;
```

→ Xiaomi 3.72★ > Apple 3.71★ > Samsung 3.70★ > OnePlus 3.68★ > Vivo 3.65★

06 BUSINESS RECOMMENDATIONS

1. Prioritise Delhi & Mumbai for All Investment

These two cities generate 43% of revenue from just 2 of 19 markets. Dedicated inventory buffers, faster delivery SLAs, and city-targeted promotions here will yield the highest return on any geographic investment decision.

2. Investigate the 2024 Revenue Decline Urgently

A –23% YoY decline is a material signal. SQL Q4 isolates the year-level trend. The next step is a brand × city × month drill-down to determine whether the decline is localised to specific markets or brands, or systemic across the business.

3. Capitalise on the July Peak Demand Window

July is the highest-revenue month consistently across all 4 years. Pre-loading inventory in the final week of June and activating promotional offers from July 1 can materially lift revenue by capturing demand at its natural peak.

4. Incentivise Digital Payment Adoption

UPI and card payments account for ~75% of transactions. Cashback offers and loyalty points for digital payments will reduce cash-handling costs, improve data traceability, and increase repeat purchase frequency.

5. Double Down on iPhone SE & OnePlus Nord

The top two SKUs contribute ■11.9 Cr combined (15.4% of total revenue). Prioritising display placement, EMI bundling, and accessory cross-sells for these two models will drive outsized revenue per customer visit across all cities.

github.com/Ajay221301/Mobile_sales_dash	Ajay Kumar Yadav | ajay172608@gmail.com	MNNIT Allahabad | May 2026